← Selected work(/#projects)

Collaborative research · 2021

Automated Psychometrics

R Shiny software for assessment analysis, supported by a co-authored PLOS ONE paper and public source code.

R · Shiny · Rasch Analysis · Psychometrics

Problem

Assessment teams need to examine test quality and group differences without writing every analysis from scratch.

Approach

Contributed to collaborative R Shiny software and research covering Rasch analysis, item bias, equating and group effects.

Kevin C. T. Chang is a co-author, credited for methodology, software and visualisation among other contributions. This was collaborative research, separate from ASB employment.

Trade-offs

Guided reports improve access, but interpretation still depends on valid measurement assumptions. The paper reports memory limits for larger online analyses.

Evidence

- Peer-reviewed paper · PLOS ONE (2021) (https://doi.org/10.1371/journal.pone.0257682)
- Public source code · AutoPsychShiny (https://github.com/kcha193/AutoPsychShiny)
- Application · autopsych (https://autopsych.shinyapps.io/version_1_0_0/)

Outcome

A documented open-source tool and peer-reviewed methods paper, with methods and source code available for inspection.

Lessons

Accessible statistical software should expose assumptions and support interpretation alongside its numerical output.

Explore the research record (/#publications) or discuss the work (/#contact).